Youxing Kong

Curriculum Vitae

Scientific AI for Geotechnical & Geohazard Systems

PhD Researcher in Geological Engineering | Beijing, China

kongyouxing@126.com

My research focuses on scientific AI for complex geotechnical and geohazard systems: learning system states from sparse, heterogeneous and multiscale observations, integrating physical principles and domain knowledge to understand evolution, and investigating failure forecasting, uncertainty analysis and trustworthy engineering decisions.

Observe → Understand → Predict → Decide

Field × Mechanics × AI

Education

2025.09 – present

PhD Researcher in Geological Engineering

Institute of Geology and Geophysics, Chinese Academy of Sciences / University of Chinese Academy of Sciences.

Research focuses on scientific AI for complex geotechnical and geohazard systems, including spatiotemporal learning, physics- and knowledge-guided modeling, long-term deformation forecasting and failure forecasting.

2021 – 2024

Master of Engineering in Geological Engineering

Institute of Geology and Geophysics, Chinese Academy of Sciences / University of Chinese Academy of Sciences.

GPA 3.66. Thesis: Mechanisms of Landslide Hazards in Fissured Expansive Soils.

Investigated fissure propagation and landslide instability through fracture-mechanics experiments, DIC and resistivity monitoring, microstructural characterization, slope model tests and field-based mechanism analysis.

2017 – 2021

Bachelor of Engineering in Civil Engineering

China University of Geosciences (Beijing).

GPA 3.68. Ranked 2 of 78 students (top 3%).

Professional Experience

2024.07 – present

Research and Engineering

Urban Safety Research Institute, Beijing Urban Construction Exploration & Surveying Design Research Institute Co., Ltd.

Work on urban infrastructure safety monitoring, intelligent inspection and research, including mobile scanning and surface-defect inspection of metro tunnels, building structural monitoring and municipal pipeline surveys.

Apply image recognition, generative image processing and time-series forecasting to engineering monitoring data, and contribute to research papers, technical reports and AI-related project proposals.

2023.10 – present

Co-founder, Kulianggong Studio

Co-founded and operate an AI-driven research and knowledge studio, exploring generative AI for research support, information synthesis, data analysis, scientific communication and knowledge management.

Design workflows from task analysis and AI workflow development to delivery, exploring lightweight, AI-assisted knowledge services and professional practices.

Research Experience

2020 – 2024

Multiscale Deformation, Failure and Landslide Instability in Fissured Expansive Soils

Investigated how fissures affect the strength, deformation and landslide instability of expansive soils. Combined fracture-mechanics experiments, DIC/resistivity monitoring, microstructural characterization and slope model testing to explain fissure propagation and instability across scales.

2024 – 2025

Tunnel Surface-Defect Recognition from Engineering Monitoring Data

Worked on deep-learning recognition, generative image completion and time-series forecasting using tunnel surface-defect and large-structure monitoring data, exploring state extraction, anomaly recognition and evolution forecasting.

2025 – present

Spatiotemporal Slope Deformation Learning and Long-term Forecasting from Sparse Observations

With STAC-GNSS as a core project, study how sparse GNSS networks can support learning of continuous slope deformation fields and evolution over multiple timescales, with further work on multisource fusion, spatial generalization and long-term forecasting.

2026 – present

Physics- and Knowledge-guided Scientific AI for Complex Geosystems

Explore the integration of data, physics and knowledge for geosystem evolution and failure, extending data-driven learning toward physical constraints, multifidelity methods, uncertainty quantification and knowledge-augmented reasoning.

Engineering and Field Practice

2020.07 – 2020.09

Atmospheric Deposition Monitoring and Analysis

Shandong Institute of Geophysical and Geochemical Exploration. Participated in field monitoring, sample collection and analysis of atmospheric deposition, gaining experience in environmental geochemical surveys, sample quality control and environmental data acquisition.

2021.09 – 2021.12

MICP Ecological Slope Stabilization Demonstration, Chuxiong, Yunnan

Participated in a demonstration using phosphogypsum resource recovery and microbially induced carbonate precipitation (MICP) for ecological slope stabilization, including microbial cultivation, material mixing, mechanical hydroseeding, planting and site construction.

2021 – 2024

Roadside Geohazard Investigation and Assessment, Second Tibetan Plateau Scientific Expedition

Combined remote-sensing interpretation with field surveys of landslides, rockfalls and debris flows along more than 2,000 km of roads. Identified 232 hazard sites and contributed to regional risk zoning.

2023.03 – 2023.06

Coal Gangue as Roadbed Fill, Yelu Expressway, Henan

Conducted compaction, vibratory compaction and California bearing ratio (CBR) tests on coal gangue with different particle sizes and gradations, and evaluated its suitability as roadbed fill using a trial section.

2024.08

Unloading Monitoring of the Large Steel Dome at Zhangjiawan Depot

Participated in strain-sensor deployment on key members, continuous stress and temperature monitoring, and data analysis to support safety control during staged unloading.

2024.10 – 2025.08

Mobile Scanning and Surface-Defect Monitoring of Urban Rail Tunnels

Participated in mobile laser scanning, seepage and surface-defect inspection, and point-cloud/grayscale-image processing. Analyzed cross-sections, convergence, segment dislocation and clearance infringement.

2025.04

Fire-door Inspection and Surveying in Metro Cross-passages

Participated in on-site fire-door inspection, engineering surveying and condition-data collection during operational maintenance windows and night shifts, gaining experience in inspection and field coordination in complex underground spaces.

Publications

Published

Wang, S., Li, Z., Li, L., **Kong, Y.**, Su, W., Qi, Z., Hu, R., & Qi, S. (2026). A nonlinear viscoelastic-plastic creep damage model of frozen soil-rock mixture under different temperatures and block conditions: Unique sights from the resistivity perspective. Engineering Geology, 370, 108831.
https://doi.org/10.1016/j.enggeo.2026.108831

Sun, K., Li, Z., Wang, H., **Kong, Y.**, Wang, S., Zhou, Y., & Hu, R. (2025). Integrated Machine Learning Using Autoencoder and Multilayer Perceptron Approaches to Predict Stress-Strain Curves of Frozen Soil-Rock Mixtures. International Journal of Geomechanics, 25(10).
https://doi.org/10.1061/IJGNAI.GMENG-10984

孔佑兴，唐超*，等 (2025). 基于 LSTM-季节性差分的钢结构穹顶卸载应力动态预测. 《测绘通报》.
https://doi.org/10.13474/j.cnki.11-2246.2025.S121

孔佑兴，唐超*，等 (2025). 基于深度学习的隧道断面扫描灰度图像渗水病害识别研究. 《工程勘察》.

Wang, Z., Wang, Y., Hou, Y., Wang, X., Chen, Y., & **Kong, Y.** (2025). 3D laser scanning combined with BIM model for construction quality inspection of large-span steel structures—A case study of the dome steel structure in Zhangjiawan Depot Comprehensive Utilization Project. In D. Bassir (Ed.), Smart manufacturing, structural health monitoring and digital twins (pp. 122–131). IOS Press.
https://doi.org/10.3233/ATDE251181

Kong, Y., Li, Z.*, Sun, K., Qi, Z., & Hu, R. (2024). Study on the extension mechanism of fissures in expansive soil based on fracture mechanics test method. Engineering Geology, 343, 107778.
https://doi.org/10.1016/j.enggeo.2024.107778

Qi, Z., Li, Z., Li, L., **Kong, Y.**, He, Z., & Zhou, Y. (2024). Study on the tensile characteristics of Brazilian split in remolded loess based on resistivity and DIC techniques. Environmental Earth Sciences, 83, 377.
https://doi.org/10.1007/s12665-024-11690-3

Sun, K., Li, Z., **Kong, Y.**, Zhou, Y., & Wang, S. (2023). Study of dynamic elastic modulus and cumulative plastic deformation in frozen soil-rock mixtures under uniaxial cyclic loading. Journal of Glaciology and Geocryology, 45(6), 1730–1743. (in Chinese)

https://doi.org/10.7522/j.issn.1000-0240.2023.0131

Li, Z., **Kong, Y.**, Fu, L., Zhou, Y., Qian, Z., & Hu, R. (2021). Model Test Study on Deformation Characteristics of a Fissured Expansive Soil Slope Subjected to Loading and Irrigation. Applied Sciences, 11(22), 10891.

https://doi.org/10.3390/app112210891

Preprints

Kong, Y., Li, Z.*, Qi, Z., Sun, K., & Hu, R. (2024). Study on Sliding Surface Characteristics of Expansive Soil Based on Quantitative Evaluation of Microstructure. Research Square.

https://doi.org/10.21203/rs.3.rs-4275076/v1

Selected Awards

2025

Outstanding Paper Award

10th Conference on Rail Transit Surveying and Exploration Technology.

Scholarships

National Scholarship; first-class academic scholarships on seven occasions.

Mathematical Modeling Competitions

Second prize in the China Undergraduate Mathematical Contest in Modeling; Honorable Mention in the Mathematical Contest in Modeling (MCM).

Research Methods and Tools

AI and Computing

Python, C++, MATLAB; graph neural networks, spatiotemporal learning, deep learning and computer vision, LLM/KG-RAG, physically constrained and probabilistic modeling.

Geotechnical Experiments and Mechanisms

Basic soil and rock tests, direct shear and dynamic triaxial tests, slope model testing; XRD, SEM/EDX, MIP, BET, DIC, resistivity monitoring and molecular simulation.

Field and Engineering

Geohazard field investigation; tunnel laser scanning and point-cloud/image processing; structural health monitoring; ArcGIS, CAD, GeoStudio/ANSYS.

Science Communication and Languages

Science Communication

Wrote 20 engineering-related popular-science articles for official Chinese Academy of Sciences platforms, with more than 100,000 total reads.

Languages

Chinese: native. English: able to conduct academic reading, writing and communication.